

DIFFERENTIAL EQUATIONS LECTURE NOTES

WEEK III

MODELLING WITH FIRST ORDER ODE'S.

(Population Dynamics)

Ex A certain bacteria population increases with a rate to the present population

- (a) If the size of population doubles in 4 hours, then what is the size of the population after 12 hours.
- (b) If there are 10^4 numbers of bacteria after 3 hours, $4 \cdot 10^4$ numbers of bacteria after 5 hours and there are $4 \cdot 10^4$ bacteria after 5 hours, then what is the number of bacteria beginning

(a) $y(t)$: size of population (number of bacteria at time t) $y(0) = y_0$
↳ initial popn

$$\frac{dy}{dt} = ky \quad y(t) = y_0 \cdot e^{kt}$$

$$t = 4 \Rightarrow y(4) = 2 \cdot y_0 = y_0 \cdot e^{4k} \quad e^{4k} = 2 \quad k$$

$$y(t) = y_0 e^{(\ln 2 / 4)t} \quad y(12) = y_0 \cdot e^{\frac{\ln 2}{4} \cdot 12} = y_0 \cdot e^{3 \ln 2} = 8 y_0$$

(b) $y(3) = 10^4$ $\frac{4 \cdot 10^4}{10^4} = \frac{y_0 \cdot e^{5k}}{y_0 \cdot e^{3k}}$ $4 = e^{2k} \quad k = \ln 2$

$y(5) = 4 \cdot 10^4$

$$10^4 = y_0 \cdot e^{3 \ln 2} \quad \boxed{y_0 = \frac{10^4}{8}}$$

Ex A certain radioactive material is known to decay at a rate proportional to the amount present. Half of this material have undergone disintegration in 50 years
 - emerged from the nucleus - particles

(a) How much material remain after 150 years?

(b) In how many years, will only 1/10 of the original material remain?

(a) $x(t)$: the amount of the radioactive material at time t .

$\frac{dx}{dt}$: the rate of radioactive decay

Decay is proportional to the amount of present $\frac{dx}{dt} = kx$

constant of proportionality $\downarrow$ by a

To emphasize that x decreases, we replace k by $-a$
 positive constant preceded by a minus sign

$$\frac{dx}{dt} = -kx$$

$$x(0) = x_0$$

$$\int_{x_0}^x \frac{dx}{x} = \int_0^t -k dt$$

$$\ln x - \ln x_0 = -k(t-0)$$

$$\ln \frac{x(t)}{x_0} = -kt$$

$$x(t) = x_0 e^{-kt}$$

$$x(50) = \frac{x_0}{2}$$

$$x_0 e^{-50k} = \frac{x_0}{2}$$

$$-50k = \ln 1/2$$

$$k = \ln 2 / 50$$

$$x(150) = x_0 e^{-(\ln 2 / 50) \cdot 150} = x_0 e^{-3 \ln 2} = \frac{x_0}{8}$$

(b) $x(t) = \frac{x_0}{10}$

$$\frac{x_0}{10} = x_0 e^{-\ln 2 / 50 \cdot t}$$

$$t = 50 \cdot \frac{\ln 10}{\ln 2}$$

Ex (Newton's law of cooling)

Experiments show that the temperature of an object changes at a rate proportional to the difference in temperature between the medium and the body itself

Suppose that a thermometer which have been showing 22°C inside a house is placed outside where the temperature is 10°C . Two minutes later it is found thermometer reading is 16°C . We wish to predict the thermometer reading at any time

$S(t)$: the temperature of the thermometer at time t .

$$S(0) = 22 \quad S(2) = 16$$

According to the Newton law the rate of change of temperature

ds/dt is proportional to $(s - 10)$ $\hookrightarrow$ Temperature of surroundings

$$\frac{ds}{dt} = -k(s - 10) \Rightarrow S(0) = 22 \quad S(2) = 16$$

$$\frac{ds}{s-10} = -k dt$$

$$\ln(s-10) = -kt + C$$

$$s-10 = e^{-kt+C} = e^{-kt} \cdot e^C$$

$$S(t) = 10 + C \cdot e^{-kt}$$

$$S(0) = 22 \Rightarrow S(0) = 10 + C \cdot e^{-k \cdot 0} = 22 \quad C \cdot e^{-k \cdot 0} = 12$$
$$C = 12$$

$$S(t) = 10 + 12 \cdot e^{-kt}$$

$$S(2) = 16 \Rightarrow 16 = 12 \cdot e^{-k \cdot 2} + 10 \quad 6 = 12 \cdot e^{-2k}$$

$$e^{-2k} = \frac{1}{2}$$

$$-2k = \ln(1/2) \quad k = -\frac{1}{2} \ln(1/2) \quad k = \ln \sqrt{2}$$

$$S(t) = 10 + 12 \cdot e^{-(\ln \sqrt{2})t}$$
$$S(t) = 10 + 12 \left(\frac{1}{\sqrt{2}} \right)^t$$

Ex (Newton's second Law of Motion)

An object of mass M is dropped from rest in a medium which offers ^(linear) resistance proportional to the velocity v of the object. Assuming the gravitational force to be constant find the velocity and the position of the object at time t .

Newton's 2nd law can be applied to this object $F = ma$
 $\downarrow$ force $\downarrow$ mass $\downarrow$ acceleration



m : mass of object g : gravitational constant

mg : force due to gravity

kv : air resistance (drag) force

k : air resistance (drag) coefficient

g : gravitational acceleration

Net force $= mg - kv$
 $\downarrow$
 F

Newton's second law $F = m \cdot a$ $\rightarrow$ acceleration $a = \frac{dv}{dt}$

$$F = mg - kv$$

$\downarrow$
 $m \cdot a$

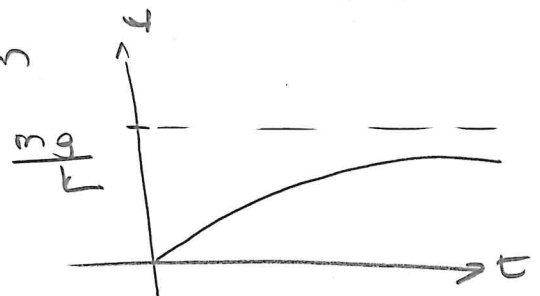
$$m \cdot \frac{dv}{dt} = mg - kv$$

$$m \frac{dv}{dt} = mg - kv$$

$$\frac{dv}{dt} = g - \frac{k}{m}v$$

$v = \frac{dx}{dt} \rightarrow$ position of object
 $\downarrow$
 exercise

Solve $v(t) = \frac{mg}{k} - \frac{mg}{k} e^{-kt/m}$



$$\frac{dx}{dt} = \frac{mg}{k} - \frac{mg}{k} e^{-kt/m}$$

Find position of vector
 $\downarrow$
 $x(t) = x_0$
 exercise

Ex 7

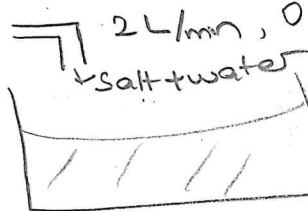
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Out): amount of salt in tank at time t

$$g(0) = 0$$

$\frac{d\sigma}{dt}$ = rate of change of salt in the tank

The rate of change of salt = $\frac{dQ}{dt} = \text{Rate in} - \text{Rate out}$

$\downarrow$

$24 \cdot 0.5 \text{ kg/L}$

$Q(t)$

$\rightarrow 24/\text{min} \times \text{concn of salt in to}$

Concentration of salt in term = $\frac{\text{amount of salt}}{\text{total amount of mixture}}$

→ 100 → since the
same amount
of mixture
gargling and
pump out

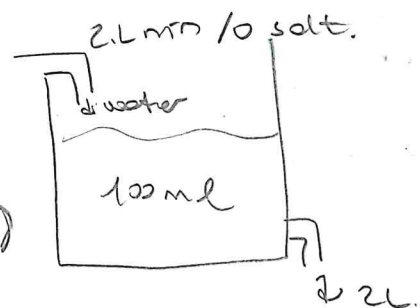
$$\left. \begin{aligned} \frac{d\theta}{dt} &= 1 - 2 \cdot \frac{\theta(t)}{100} \\ \theta(0) &= 0 \end{aligned} \right\} \text{I.V.P}$$

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$$\textcircled{b} Q(t) = 50 (1 - e^{-t/50})$$

$$t = 10 \quad Q(10) = 50 (1 - e^{-1/5})$$

$$R(10) = Q(10) = 50 (1 - e^{-1/5})$$



$$\begin{aligned} \frac{dQ}{dt} &= \text{Rate in} - \text{Rate Out} \\ &= 2.0 - 2 \cdot \frac{Q(t)}{100} \end{aligned} \quad \left. \begin{array}{l} \text{I.V.P} \\ \Rightarrow \end{array} \right\}$$

$$\begin{aligned} \frac{dQ}{dt} &= -\frac{Q(t)}{50} \\ Q(10) &= 50 (1 - e^{-1/5}) \end{aligned}$$

0 salt.

When we solve we obtain

$$Q(t) = 50 (1 - e^{-t/50}) e^{-1/50 t}$$

$$\text{as } t \rightarrow \infty \quad Q(t) \rightarrow 0$$